# JIANT FM · SOUND · FX · MOD

engines, effects, modulation, sequencer and songs

firmware v0.9.8 · M-VAVE FM-1 · juanjiant-bit.github.io/JIANT

## ENGINES · EDIT 1 / EDIT 2

### NINE ENGINES, ALL SYNTHESIS

| ENGINE | EDIT 1 | | | | EDIT 2 | | | | WHAT TO KNOW |
|---|---|---|---|---|---|---|---|---|---|
| ANALOG | WAVE | DETUNE | MIX | NOISE | CUT | RES | DRIVE | INT | SAW SQR TRI SIN PWM **SYNC RING SAW3**; INT: osc 2 ±24 st. Its filter is per voice. **Its picture** (0.9.6): a cell whose membrane is the wave, spikes where DRIVE clips, a ring at CUT. |
| FM6 | ALG | FB | MOD | RATIO | MEG | VMOD | DETUNE | SLOT | Dexed's 6 operators, 32 algorithms drawn, .syx patches (SLOT), MOD / RATIO move the modulators. **OPERATOR · OP ENV · OPERATOR 2**: each operator on the device (ratio / fixed, fine, level, its envelope, mode, detune, velocity). **SLOT B1–B32**: a 32-voice .syx sent from the web editor. **Its picture** (0.9.6): the operators as bichos wired as the algorithm, OP ENV's riding its envelope. |
| PHASE | WAVE | WAVE2 | DCW | ENV | DETUNE | LINE | SUB | FB | Phase distortion (CZ); **FB** feeds the output back into the phase: hard edge to growl. **Its picture** (0.9.6): a snail, its shell's chambers at the bent phase, its sole the wave. |
| LOFI | CHIP | WAVE | DUTY | CRUSH | - | VIB | BEND | MASK | Chip: PLS TRI SAW NOIS WRAM. **BYTE**: ALGO VAR · MOTN GRIT BEND LOOP. **FLOAT** (floatbeat): ALGO VAR · MOTN FBK BEND LOOP; CHIP sets FULL / 4BIT / 1BIT. **Its picture** (0.9.6): a caterpillar whose skin is the output, a segment per bit of the byte. |
| VOICE | VOWEL | VOWEL2 | TALK | SHIFT | BUZZ | BREATH | Q | RAND | Formant voice (after klattsch): vowels, talk, breath. **WORDS** (0.8.5): LANG ES / EN / JA sings syllables, a note each (POS THEME ARTIC SHIFT · BUZZ BREATH Q LOOP); LYRIC TEXT (OCT+ on WORDS types it, KNOB 3 dice; the web editor sends it) or W1–16 words, S1–8 scat; "-" holds the vowel. **Its picture** (0.9.4): a CT head whose tongue, jaw, lips and velum sing the phoneme. |
| WHEEL | REG | SUB | BODY | TOP | PERC | CLICK | DRIVE | ROTOR | Drawbar organ: registrations, percussion, rotary. **Its picture** (0.9.6): nine tube worms, one a drawbar, bent by the rotor. |
| NOISE | MODE | COLOR | FREQ | PEAK | TRK | DENSE | DRIFT | CRUSH | Coloured and metallic noise, tracked or free. **Its picture** (0.9.6): a swarm walking by COLOR, swirling by FREQ. |
| DRUM-X | MORPH | TUNE | TONE | DECAY | NOISE<br>DRIVE | FM | ENV | FOLD | 8 sounds BD SD CP CH OH TM RS CB, sides A / B: **MORPH** morphs the kit. The kit first: **KIT 3** BP FEEDBK, **X-MOD** RATE DEPTH SHAPE RAND. Then a sound (KNOB 1 the sound): SIDE PITCH PMOD · DECAY NOISE COLOR · MODE PM.TYPE DRIVE · **LAYER** (no note) FEEDBK **SEND** (CHO / DLY / REV). Last LEVELS and **STEREO** IMAGE (a knob through the kit's stereo placings). KIT, SOUND 1–4 and HOME › BEING draw the kit as little beings that squash and shout on each hit (0.9.4). |
| ANIMAL | MODE | PRESS | CHAOS | SIZE | SWEEP | RATE | GROUP | TEMP | Animal calls (0.8.3), MODE the source: **CMOS** the 40106 / 4093 bird circuit (TOUCH COUPLE CAP · HYST RATE GROUP FM) · **PULSE** crickets, cicadas, frogs (CLICK Q · DETUNE) · **WHSTL** birds, a Hopf whistle · **WING** mosquito, bee, fly (BUZZ DRIFT · DOPPLR) · **GLOTT** wolf, bark, roar, purr (MOUTH ROUGH · AIR). GROUP: syllables + gaps; TEMP °C speeds a cricket up (Dolbear). A note is a call. **Its picture** (0.9.4): bird ↔ insect ↔ mammal by MODE, leaned by SIZE. |

### FILTER · EVERY ENGINE

| | | | |
|---|---|---|---|
| TYPE | CUT | RES | DRV |

**LP · HP · BP · COMB**. COMB rings tuned to the note: CUT moves it ±32 st, RES how much it rings. ANALOG and LOFI: per voice; the others: on the track, following LFO / ENV DEST FLT (DRV = the track's DIST).

### VOICE · PLAYING

| | | | |
|---|---|---|---|
| VCE | GLD | GLMOD | PRIO |

POLY · MONO · LEG · UNI; glide and its mode; LAST / LOW / HIGH priority. **8 voices** shared by the six tracks (MONO and LEGATO keep their note).

### DICE

**EDIT + SELECT**: a random factory preset of the engine, every parameter moved up to a quarter of its range. DRUM-X: both sides of every sound around the factory kit. Heard at once: OCT+ OK keeps it, OCT- CANCEL brings the sound back. SEQ / LFO / FX + SELECT: the sequence, the LFO, the effects, the same way. HOME + SELECT (or HOME twice on HOME): **GEN**, the song generator, HOME's last three pages: GENRE (**EXPLORE**, where it opens: no genre, every rule broken; RANDOM; 43 genres; PRESETS too) · SEED · ENERGY · FORM; DRUMS · BASS · CHORDS · ARP · LEAD amounts; COMPLEX · SWING · HUMAN (SELECT: the next page). OCT+ makes the piece (again: a new seed), OCT- back to ECO. GLO + G4 undoes.

## MODULATION

### ENV · LFO

ENV

| | | | |
|---|---|---|---|
| ATK | DEC | SUS | REL |

ENV DEST

| | | | |
|---|---|---|---|
| FLT | PIT | SHP | LOOP |

LOOP: held, the envelope cycles from the sustain: an ADSR-shaped LFO.

LFO

| | | | |
|---|---|---|---|
| RATE | WAVE | PHS | FADE |

SIN TRI SAW SQR S&H

LFO DEST

| | | | |
|---|---|---|---|
| PIT | FLT | SHP | AMP |

### MOD MATRIX · 4 SLOTS A TRACK

| | | | |
|---|---|---|---|
| SLOT | SRC | DST | AMT |

SOURCES

LFO ENV VEL KEY RAND MODW AT EXPR M1 M2 M3 M4 STEP

DESTINATIONS

PITCH CUT SHP AMP PAN DIST CHO DLY REV RATE VIB E1...E8 CLIP PNCH D.TIM D.FB D.TON D.MIX R.SIZ R.DMP C.RAT C.DEP

E1...E8: the engine's own EDIT knobs. CLIP / PNCH: the master. D. R. C.: the DELAY, REVERB and CHORUS buses (a STEP source on D.TIM sequences the delay time).

### MSEQ · MACROS

MSEQ · A CV SEQUENCER PER TRACK

| | | | |
|---|---|---|---|
| STEP | LVL | LEN | SLEW |

16 levels, DIV on MSEQ 2, in time with the transport. Use it from MOD, SRC STEP.

MACROS M1–M4

Hold LFO: the knobs are M1–M4, latched until LFO again. Each session starts with 2 random routes a track; F3 rolls new ones, G3 clears. By hand: MOD, SRC M1...M4. Saved with the project.

### SEQUENCER

64 steps a track, up to 4 notes a step, piano roll and drum grid. One chain, five groups: PATTERN · TRACK · BAR (BAR: K1 the bar; STEP: the roll / grid, K1 all 64 steps; NOTE) · STEP (COMP: one step large, its chips) · AUTO; SELECT walks, SEQ jumps to the next group, PRESETS loads a pattern. **Hold a step + turn a knob**: a parameter lock. Step recording: hold a chord + KNOB 1 its length; + ARP: arpeggiated (K1 rate, K2 order, K3 oct). COMP / COMP 2: two components a step (CHANCE, RATCH, ARP NOTES / SEQ / SCALE, TRIG, RAND, OCT, TONAL, BEND, ECHO, PULSE, JUMP, REV, FILL, KILL, WALK, CLIMB, TOUCH: a kit's SHORT / GHOST / FLAM); a TRIG or FILL wins; WALK / CLIMB: a degree / an octave further every N loops (K4), UP DOWN UP-DN RND over a range (K3), stackable; ARP: ORDER × range, RATE in note values (x1 1/32 1/32T 1/64 1/64T 1/128); RATCH / ECHO: DIV; EVERY (KNOB 4): the loops the step's components play on (ALWAYS 1:2 2:2 1:3 .. 4:4 1:8 FIRST !FIRST SKIP). TRACK: PITCH, CHANCE thins the whole track, HUMAN one amount (velocity, timing, gate, decay / release), REC Q. STRETCH / REPEAT act as the knob turns; ROTATE reads the sequence from another step; NOTE's MICRO moves a step off the grid in 64ths (±32/64; REC Q 0 % records where you played, to the 64th). REC + knobs while playing: automation (64 records a section, MOTION page plays / clears them, AUTO LIST edits them).

PATTERN · TRACK

| | | | |
|---|---|---|---|
| LEN | DIV | SWG | GATE |
| PIT | CHN | HUM | RECQ |

### SCALE · ARP

**ROOT and SCALE** are every melodic track's: CHR MAJ MIN DOR MIX PEN MPEN HARM PHRY LYD LOC MEL BLUES WHOLE DIMHW DIMWH. QNT snaps keys or the sequence, **ALL TR** (SCL KNOB 4) moves every sequence along the scale. **CHORD**: a key plays a chord (TRIAD 7TH ADD9 SUS2 SUS4 in key, MAJ MIN 7 MAJ7 MIN7 POWER), VOICING LEAD leads each to the next, STRUM as a harp.

ARP MODES

OFF UP DN UPDN RND ORD REPEAT DNUP UP+8 CONV DIVG PINKY THUMB WALK CHORD **TRNS** (the keys transpose the sequence)

### SONGS

**8 projects × 4 sections (A–D)**, a song of up to 16 rows by bars (SAVE › PROJECT › ARRANGE, OCT+ adds a row, PLAY plays; END: STOP, x2–x8, LOOP), SLOOP style. Each row can hold a **scene**: mutes, DRUM-X group mutes, macros and MIDI punch-ins. SAVE held plays sections live on the next bar, stores them, records the song (SONG REC). Each section keeps its sounds, steps, motion, kit and its SLICER.

## EFFECTS

### PER TRACK

FX · SENDS

| | | | |
|---|---|---|---|
| DST | CHO | DLY | REV |

DIST

| | | | |
|---|---|---|---|
| DST | TYPE | TONE | - |

SOFT · HARD · FOLD · CRUSH · RECT.

AUTO PAN

| | | | |
|---|---|---|---|
| DEPTH | RATE | SYNC | WAVE |

The track's pan swung L–R around its PAN. SYNC **BEAT**: RATE 4 BARS ... 1/32, in time with the bar; **FREE**: in Hz. SIN · TRI · SAW · SQR · S&H, never a click.

### SLICER · ONE ON A BUS

| | | | |
|---|---|---|---|
| SLCR | PAT | RATE | DEPTH |

OFF · GATE · STUT, 16 patterns, 1/8 ... 32T. **SL BUS**: T1–T6 and **FX** (the delay / reverb / chorus returns) ON / OFF: everything on = the whole mix chopped after the effects. **SL PITCH**: the repeats' 16-step pitch (±24 st, tape style, QNT SCL to the scale).

### DELAY · CHORUS

DLY

| | | | |
|---|---|---|---|
| TIME | FDBK | TONE | MIX |

DLY 2

| | | | |
|---|---|---|---|
| WIDE | PITCH | SPRY | MOD |

DLY 3

| | | | |
|---|---|---|---|
| TYPE | PING | - | - |

Synced or free (5 ms – 2.95 s; past 0.74 s darker); PITCH / SPRY: a grain delay.

CHORUS

| | | | |
|---|---|---|---|
| RATE | DEPTH | STER | - |

STER: 0 as always, down to mono, up to twice as wide.

### REVERB · FIVE MODELS

| | | | |
|---|---|---|---|
| TYPE | SIZE | DAMP | PRE |
| MOD | RATE | FILT | WIDE |

| TYPE | WHAT IT IS | SIZE | DAMP | MOD | RATE | WIDE |
|---|---|---|---|---|---|---|
| ROOM | Classic room | decay | darkens | chorus on the tail | its speed | stereo |
| SPRING | Amp spring tank | length · decay | darkens | wobble | wobble speed | stereo |
| SHIMR | Up an octave on every pass; **PRE → AIR** (±24 st: down darker and lower, up airier and higher) | decay | darkens | how much shimmer | — | — |
| RESO | 4 strings tuned to the scale; **PRE → PITCH** (0–14 steps up), **DAMP → FOLW** (OFF / T1–T6: the chord follows that track's notes) | how long they ring | FOLW | past half: the chord opens | — | — |
| CLOUD | Granular wash; SIZE max **freezes** it; **PRE → AIR** (±24 st: the cloud down / up, darker / airier) | decay · freeze | darkens | grains pitched + chorus | grain density | grain spread |

### MASTER

| | | | |
|---|---|---|---|
| CLIP | PNCH | DUCK | REL |

**CLIP** saturates without getting louder. **PNCH** drives the drums: denser body. **DUCK**: the kick pulls the rest down (REL its release). An always-on **leveler** (slow 2:1, −9...+6 dB) before the limiter keeps patches at a steady volume.

PUNCH-IN FX (FX HELD)

REPEAT 1/8 · 1/16 · 1/32, LPF, HPF; on the notes: OCT–/+, ½ TEMPO, DEC–/+, STUTTER, ATK+, ARP, RANDOM. Knobs: FILTER CRUSH THROW DEPTH. Recordable per section.

GPL-3.0 · fork of Felucca (Hügelton Instruments) · song mode from SLOOP